

Stefan A. Wirler

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Research Interests

Spatial Audio Processing, Microphone Array Processing, Beamforming, Audio Signal Processing, Room Acoustics Modeling, Machine Learning, Virtual Acoustics Perception

Education

Aalto University **Espoo, Finland**
D.Sc.(Tech) / PhD Acoustics and Audio Signal Processing 2020–2026

Thesis: *Spatial post-filtering for speech enhancement and source separation*
Department of Information and Communications Engineering, Acoustics Lab
Supervisor: Prof. Ville Pulkki

Friedrich-Alexander-University (FAU) **Erlangen, Germany**
M.Sc. Electrical Engineering, Grade: 1.5 – "Very Good" – 90/100 2016–2019

Thesis: *Impact of Diffuse and Disturbed Reflections on Room Geometry Inference Algorithms*
International Audio Laboratories Erlangen - Joint Institution of Fraunhofer IIS and FAU

Regensburg University of Applied Science **Regensburg, Germany**
B.Eng. Electrical Engineering, Grade: 1.8 - "Good" – 84/100 2012–2016

Thesis: *Multichannel Room Impulse Response Measurement for the Determination of Room Acoustic Parameters*
Electroacoustics Lab, Regensburg University of Applied Science

Experience

Tampere University **Tampere, Finland**
Postdoctoral Research Fellow 2026–

Rendering of six-degrees-of-freedom sound scenes from multipoint recordings for binaural reproduction.
Supervisor: Prof. Archontis Politis

AAC Technologies **Turku, Finland**
Project Worker 2021–2023

Microphone array design, implementation of beamforming algorithms and development of parametric time-frequency domain spatial post-filtering algorithms for hand-held devices (MATLAB)

Aalto University **Espoo, Finland**
Research Assistant 2020

Implementation of a real-time binaural rendering system for the evaluation of virtual acoustic perception (Max/MSP)

Fraunhofer IIS **Erlangen, Germany**
Graduate Research Assistant 2019

Implementation of FDN reverberation algorithms for the evaluation and comparison to real-world recordings (MATLAB)

FAU

Research Internship

Implementation and extension of an independent vector analysis algorithm to support block-online processing (Python)

Erlangen, Germany

2018

National Instruments Germany GmbH

Application Engineering Intern

Munich, Germany

2014–2015

Others

Aalto University

Teaching Assistant, Acoustics and Physics of Sound

Espoo, Finland

2021–2024

Aalto University

Thesis Advisor

Espoo, Finland

Master Thesis

“Real-time Implementation and Evaluation of Acoustic Occlusion in Virtual Reality”, Andrés Ortiz Pachón, 2021

“Evaluation of Pair-Wise Similarity Spotforming Algorithm on Real Omnidirectional Signals and Ambisonic Signals with Search for Improvements on the Algorithm”, Antoine Souchaud, 2024

Computer skills



MATLAB



Max/MSP



Python



Assembly



C



VHDL



C++



LabVIEW

Publications & Patents

Publications

- [1] **Wirler, Stefan**, Nils Meyer-Kahlen, and Sebastian J Schlecht. **Towards transfer-plausibility for evaluating mixed reality audio in complex scenes**. In *Audio Engineering Society Conference: 2020 AES International Conference on Audio for Virtual and Augmented Reality*. Audio Engineering Society, 2020.
- [2] **Wirler, Stefan**, Sebastian J Schlecht, and Ville Pulkki. **Machine learning based auralization of rigid sphere scattering**. In *2021 Immersive and 3D Audio: from Architecture to Automotive (I3DA)*. IEEE, 2021.
- [3] **Wirler, Stefan** and Ville Pulkki. **Spatial post-filter estimation based on low-order beam-formers**. In *International Congress on Acoustics*. Acoustical Society of Korea (ASK), 2022.
- [4] **Wirler, Stefan**, Vasileios Bountourakis, and Ville Pulkki. **Space-domain cross-pattern coherence post-filter for speech enhancement with linear microphone arrays**. In *Audio Engineering Society Convention 154*. Audio Engineering Society, 2023.
- [5] **Wirler, Stefan**, Nils Meyer-Kahlen, and Ville Pulkki. **Enhancing Spatial Post-Filters through Non-Linear Combinations**. In *Audio Engineering Society Convention 157*. Audio Engineering Society, 2024.

- [6] **Wirler, Stefan** and Pulkki Ville. **Spatially Selective Sound Capture Based on Aggregated Pairwise Similarity Measures**. *Journal of the Audio Engineering Society*, 73:747–759, Nov. 2025.
- [7] **Wirler, Stefan**, Nils Meyer-Kahlen, and Ville Pulkki. **Synthesizing a Virtual Height Channel from Planar Microphone Arrays**. In *2025 33rd European Signal Processing Conference (EUSIPCO)*. IEEE, 2025.
- [8] Nils Meyer-Kahlen, Daniel Rudrich, Manuel Brandner, **Wirler, Stefan**, Simon Windtner, and Matthias Frank. **Diy modifications for acoustically transparent headphones**. In *Audio Engineering Society Convention 148*. Audio Engineering Society, 2020.

Patents.....

- [9] **Wirler, Stefan Albert** and Juha Backman. **Microphone signalbeamforming processing method, electronic device, and non-transitory storage medium**. US Patent App. 18/625,253, October 9 2025.
- [10] **Wirler, Stefan Albert**, Juha Backman, Timo Lamberg, and Antti Kangasaho. **Microphone array and portable device**. US Patent App. 18/888,193, March 19 2026.